

Sungwoo Park

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RESEARCH INTERESTS

My research focuses on UTM and AAM for safe, efficient, and scalable operations. My research interests include decentralized traffic management, multi-agent conflict resolution, path planning, and GIS-based airspace network design. My previous research experience includes air risk analysis and ground risk analysis, providing a foundation for understanding the safety challenges associated with UAS operations. Building on these experiences, my current research focuses on game-theoretic and optimization-based approaches to managing large-scale UAS operations and designing realistic urban airspace networks.

EDUCATION

Korea Aerospace University

Master of Science – MS, Department of Advanced Air Transportation

Mar. 2025 – Present

Goyang, South Korea

- **Advisor:** Dr. Sang Hyun Kim
- **Grade:** 4.2 / 4.5
- **Key Courses:** Optimization Theory and Applications, Introduction to Advanced Artificial Neural Networks, Machine Learning with Python, Simulation-Based Disaggregate Choice Modeling

Korea Aerospace University

Bachelor of Science – BS, Air Transport, Transportation, and Logistics

Mar. 2019 – Feb. 2025

Goyang, South Korea

- **Grade:** 4.08 / 4.5
- **Focus:** Aviation Regulations, Aviation Meteorology, Airport Planning, Air Navigation Facilities and Air Navigation, Aeronautical Information, Aeronautical Communications, ATC Training 1 and 2, Operation Research 1 and 2

RESEARCH PROJECTS

- P1.** Decentralized UAS Traffic Management
Supervised by Prof. Sang Hyun Kim, 2024–2026.
- P2.** Population Estimations for Mitigating Ground Risk in Unmanned Aircraft Systems
Bachelor Thesis, Supervised by Prof. Sang Hyun Kim, 2024–2026.
- P3.** Air Risk Analysis
Supervised by Prof. Sang Hyun Kim and Lee Keumjin, 2025.
- P4.** Evaluation of Obstacle Limitation Surfaces in UAM Operations
2022 Korea Student UAM Olympiad, Supervised by Prof. Sang Hyun Kim, 2022.

PUBLICATIONS AND MANUSCRIPTS

Journal Manuscripts

- J1.** Ji, S., **Sungwoo Park**, Kim, S. H., “Risk-Based Airspace Capacity Assessment for Small Unmanned Aircraft in Urban Environments.”
Transportation Research Part A: Policy and Practice, under review, 2026.
- J2.** **Sungwoo Park**, S. H. Kim, “Game-Theoretic Approaches to Decentralized UAS Traffic Management.”
IEEE Transactions on Intelligent Transportation Systems, manuscript in preparation for submission, 2026.

Conference Publications and Presentations

- C1.** **Sungwoo Park** and S. H. Kim, “Multilateral Negotiation Among Traffic Management Service Providers.”
AIAA AVIATION 2026 Forum, available at [\[here\]](#).
- C2.** **Sungwoo Park** and S. H. Kim, “Fairness among UAS service suppliers under different negotiation rules.”
The Korean Society for Aeronautical and Space Sciences 2026 Spring Conference, available at [\[here\]](#).
- C3.** B. Kang, **Sungwoo Park** and S. H. Kim, “Analysis of Available Airspace for Drones in Airport Control Zone.”
The Korean Society for Aeronautical and Space Sciences 2025 Fall Conference, available at [\[here\]](#).
- C4.** **Sungwoo Park** and S. H. Kim, “Game-Theoretic Multi-Agent Traffic Management for Fair and Efficient Low-Altitude Operations.”
The Society for Aerospace System Engineering 2025 Spring Conference, available at [\[here\]](#).
- C5.** **Sungwoo Park**, H. J. Jin, and S. H. Kim, “Ground Risk Analysis for Urban Low Altitude Operations.”
The Korean Society for Aeronautical and Space Sciences 2024 Spring Conference, available at [\[here\]](#).

HONORS AND ACADEMIC ACTIVITIES

- H1.** First Prize, 2023 Air Transportation Capstone Design II, Korea Aerospace University.
- H2.** Third Prize, 2023 Korea Student UAM Olympiad, Korea Transport Institute.
- H3.** Second Prize, 2022 Korea Airports Corporation Student Thesis Competition, Korea Airports Corporation.

CERTIFICATIONS AND LICENSES

- L1.** Pilot Certificate for Ultra-Light Vehicles, Ministry of Land, Infrastructure and Transport, South Korea, 2022.
- L2.** Machine Learning with Python-From Linear Models to Deep Learning, edX, 2025

TECHNICAL SKILLS

Programming: Python, MATLAB, QGIS, LaTeX

Research Methods: Game theory, Optimization, Simulation

Languages: Korean (Native); English (TOEFL iBT: 5.0 / 6.0)